Partial Derivatives

For problems 1 – 13 find all the 1st order partial derivatives.

1. $f\left(x,y,z\right)=x^3\sqrt{y}+4z^3y^2-xyz+x^2-\sqrt[3]{z^5}$

2. $W\left(a,b,c,d\right)=a^2+b^3-c^2d^4-5a^3c-d^6b^2a$

3. $A\left(p,t,u\right)=\dfrac{1}{pt^2}-\dfrac{t^3}{u^2}+\dfrac{4u\,p}{t^4}$

4. $g\left(x,y,z\right)=\sqrt{x^2+z^{-2}}+\sin\left(xy-x^2\right)$

5. $f\left(s,t\right)=\cos\left(s\mathbf{e}^{t^2}\right)+\cos\left(s+\mathbf{e}^{t^2}\right)$

6. $f\left(x,y\right)=\ln\left(\dfrac{y}{x}\right)+\ln\left(\dfrac{1}{x+y}\right)-\ln\left(\dfrac{x}{6}\right)$

7. $A\left(y,z\right)=\dfrac{1}{y-4z^5}+\tan\left(yz^2-y^3\right)$

8. $g\left(u,v\right)=\dfrac{u}{v}\cos\left(\dfrac{v}{u}\right)+4u-v^2u$

9. $w=\left(x-y\right)\mathbf{e}^{4x+z^6}-\sin\left(2x+7z\right)\sec\left(yz^3\right)$

10. $f\left(u,v,w\right)=\left(uw+4\right)\sin^{-1}\left(u^2+v^2\right)-\ln\left(\dfrac{w^2}{v^4}\right)$

11. $f\left(x,y,z\right)=\sin\left(\dfrac{z}{z^2+x}\right)-\dfrac{6x^2+y}{y^2-z^2}$

12. $g\left(s,t,p\right)=\dfrac{p^3t^2}{s^2+1}+\dfrac{\left(4s-1\right)t^2}{6-s}$

13. $f\left(x,y,z,w\right)=x^2\sin\left(4y\right)+z^3\left(6x-y\right)+y^4$

For problems 14 & 15 find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the given function.

14. $z^4 - y^2 + x^2 = 6x^2y^3z^7$

15. $x^2 \sin(z) + (x^2 - 1)y^4 = z^6$